

P4 MULTISPECTRAL

Quick Start Guide

快速入门指南

快速入門指南

クイックスタートガイド

퀵 스타트 가이드

Kurzanleitung

Guia de início rápido

Guide de démarrage rapide

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Snelstartgids

Guia de início rápido

Guia de Início Rápido

Краткое руководство пользователя

v1.2

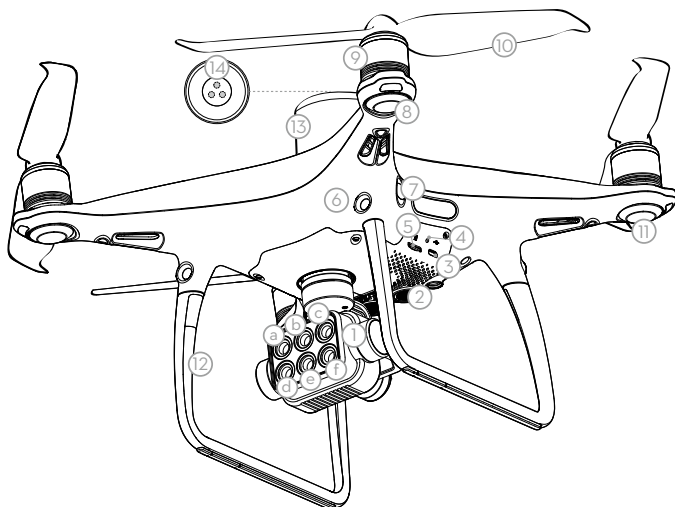


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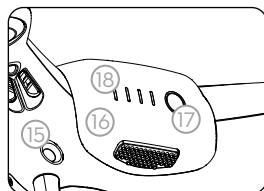
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P4 Multispectral

The P4 Multispectral is a high-precision drone capable of multispectral imaging functions. The imaging system contains six cameras with 1/2.9-inch CMOS sensors, including an RGB camera and a multispectral camera array containing five cameras, all at 2MP with global shutter, on a 3-axis stabilized gimbal. The spectral sunlight sensor on top of the aircraft detects the solar irradiance in real-time for image compensation, maximizing the accuracy of collected multispectral data. Image data can be used to generate multispectral maps for plant and soil status analysis. The aircraft has a built-in DJI™ Onboard D-RTK™, which provides precision data for centimeter-level positioning accuracy*. Multi-directional obstacle sensing is enabled by forward, rear, and downward vision and infrared sensors*.



- | | |
|---|--------------------------------|
| 1. Gimbal Cameras
(with six cameras corresponding to the wave bands below) | 8. Front LEDs |
| a. Red Edge (RE) b. Near-Infrared (NIR) | 9. Motors |
| c. Green (G) d. Visible Light (RGB) | 10. Propellers |
| e. Red (R) f. Blue (B) | 11. Aircraft Status Indicators |
| 2. Downward Vision System | 12. Ocusync™ Antennas |
| 3. Micro USB Port | 13. Onboard D-RTK Antenna |
| 4. Camera/Linking Status Indicator and Link Button | 14. Spectral Sunlight Sensor |
| 5. Camera microSD Card Slot | 15. Rear Vision System |
| 6. Forward Vision System | 16. Intelligent Flight Battery |
| 7. Infrared Sensing System | 17. Power Button |
| | 18. Battery Level Indicators |

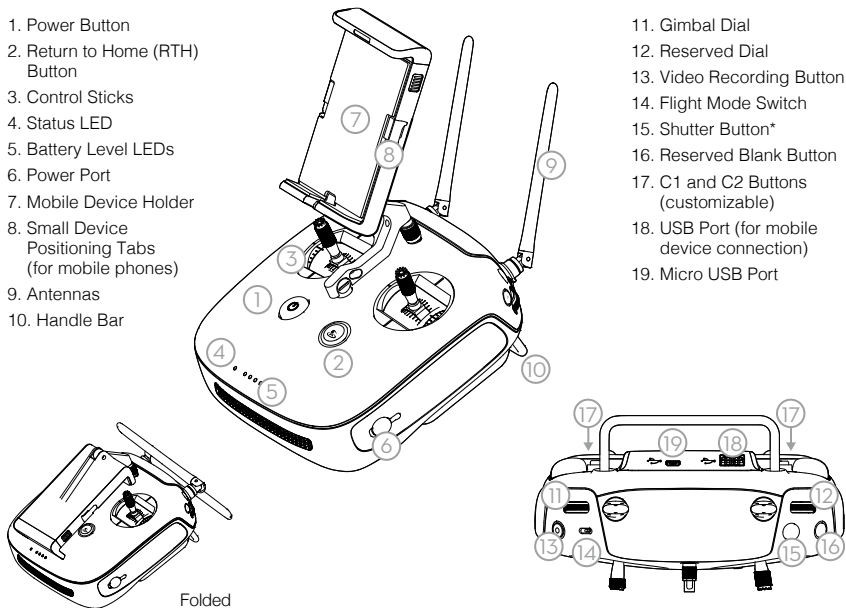


* This should be used with Network RTK service, a DJI D-RTK 2 High Precision GNSS Mobile Station (purchased additionally) or post-processed kinematic (PPK) data (recommended when RTK signal is weak during operation).

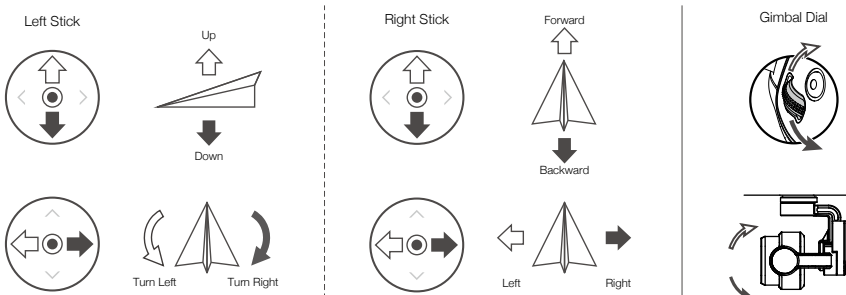
The Vision and Infrared Sensing Systems are affected by surrounding conditions. Read the User Manual to learn more.

Remote Controller

The remote controller features DJI's long-range transmission technology OcuSync that is capable of controlling the aircraft and the gimbal cameras at a maximum transmission range of 4.3 mi (7 km)*. Connect an iPad to the remote controller via the USB port to use the DJI GS Pro app to plan and perform missions. Export the captured images for analysis and create multispectral maps. The remote controller has a rechargeable LiPo battery with a maximum battery life of approximately 6 hours*.



The figure below shows the function that each control stick movement performs, using Mode 2 as an example. The left stick controls the aircraft's altitude and heading, while the right stick controls its forward, backward, left and right movements. The gimbal dial controls the camera's tilt.



* The remote controller is able to reach its maximum transmission distance (FCC/NCC) in a wide open area with no Electro-Magnetic Interference, and at an altitude of about 400 ft (120 m). The maximum runtime is tested under laboratory environment, only for your reference. Pictures will only be taken when the shutter button is fully pressed.

Using the P4 Multispectral

1. Downloading the DJI GS Pro App

The latest version of DJI GS Pro is required when using with the P4 Multispectral. Search for DJI GS Pro* in App Store or scan the QR code to download the app on your iPad.



DJI GS Pro

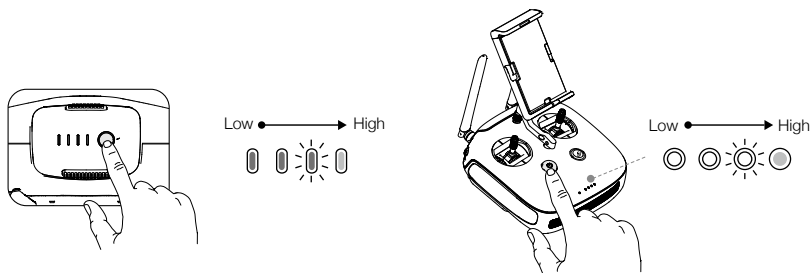


Internet

When using your P4 Multispectral for the first time, activate it using DJI GS Pro. Ensure that your iPad has access to the internet.

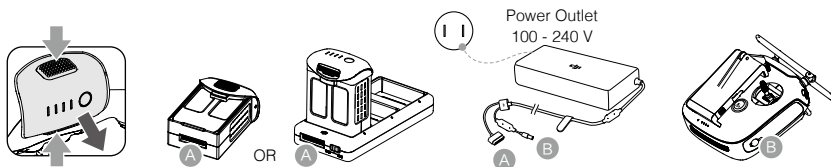
* Visit the official DJI website for more information about DJI GS Pro. <https://www.dji.com/ground-station-pro>

2. Checking Battery Levels



Press once to check the battery level. Press once, then again and hold to turn on/off.

3. Charging the Batteries



- Fully charge the batteries before using for the first time.
- Ensure to connect the Intelligent Flight Batteries to the charging hub as shown in the figure above.
- Ensure that the Mode Switch of the Intelligent Flight Battery charging hub is set to the **Charging Mode** position.

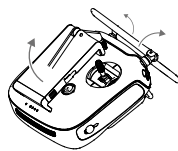
4. Preparing the Remote Controller



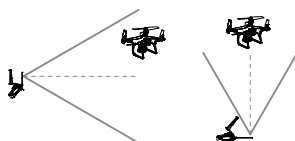
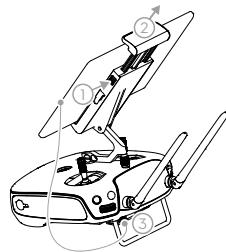
Strong



Weak



Unfolding the Remote Controller



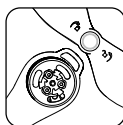
Optimal Transmission Zone

Try to keep the aircraft inside the optimal transmission zone. If the signal is weak, adjust the antennas or fly the aircraft closer.

5. Preparing for Takeoff



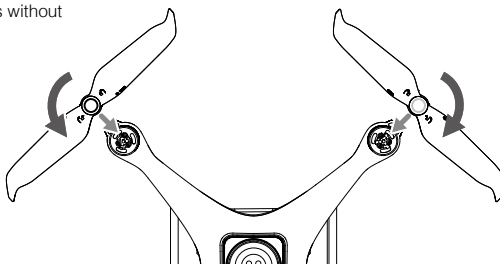
Black propeller rings go on motors with black dots.



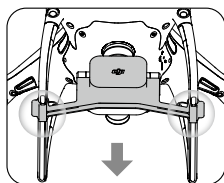
Silver propeller rings go on motors without black dots.



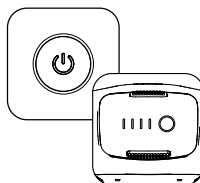
Press the propeller down onto the mounting plate and rotate in the lock direction  until secure.



• Check that the propellers are secure before each flight.



Remove the gimbal clamp from the camera.



Power on the remote controller and the aircraft.

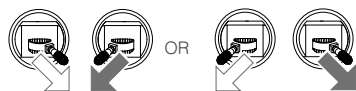


Launch the app.

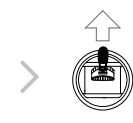
6. Flight

For a safe take-off, make sure that the Aircraft Status Indicators blink green slowly (using RTK* or GNSS for positioning).

Takeoff



Combination Stick Command to start/stop the motors



Left stick (in Mode 2) up slowly to take off

Landing



Left stick down slowly until you touch the ground
Hold for three seconds to stop the motors

- ⚠ Spinning propellers can be dangerous. Stay away from spinning propellers and motors. DO NOT start the motors in confined spaces or when there are people nearby.
- Always keep your hands on the remote controller when the motors are spinning.
- Stopping motors mid-flight: Perform the CSC to stop the motors. This function can be enabled in DJI GS Pro. Only stop motors mid-flight in emergency situations when doing so can reduce the risk of damage or injury.**

* RTK positioning is recommended. Go to Mission View in DJI GS Pro, tap the icon  or RTK on top of the screen to go to the RTK settings menu, and then select D-RTK 2 or Network RTK Account as the RTK data source. Enable the aircraft RTK at the bottom of the menu. Otherwise, the aircraft cannot use the RTK data.

7. Starting Operations

The following example includes instructions for 3D Map Area missions using DJI GS Pro.



Create a 3D Map Area mission.



Configure mission parameters.



Tap camera preview to enter Camera View to configure camera settings*.



Perform the mission.



Export the pictures.



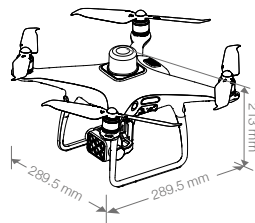
Generate a map.

* When multispectral camera has been selected in Camera View, this mission will not include RGB photos.

Specifications

• Aircraft

Takeoff Weight	1487 g
Diagonal Distance (Propellers Excluded)	350 mm
Max Service Ceiling Above Sea Level	19685 ft (6000 m)
Max Ascent Speed	6 m/s (automatic flight); 5 m/s (manual control)
Max Descent Speed	3 m/s
Max Speed	31 mph (50 kph) (P-mode); 36 mph (58 kph) (A-mode)
Max Flight Time	Approx. 27 minutes
Operating Temperature	0° to 40° C (32° to 104° F)
Operating Frequency	2.4000 GHz to 2.4835 GHz (Europe, Japan, Korea) 5.725 GHz to 5.850 GHz (Other countries/regions) ¹⁾
Transmission Power (EIRP)	2.4 GHz: < 20 dBm (CE / MIC / KCC) 5.8 GHz: < 26 dBm (FCC / SRRC / NCC)
Hover Accuracy Range	RTK enabled and functioning properly: Vertical: ±0.1 m; Horizontal: ±0.1 m RTK disabled: Vertical: ±0.1 m (with vision positioning); ±0.5 m (with GNSS positioning) Horizontal: ±0.3 m (with vision positioning); ±1.5 m (with GNSS positioning)
Image Position Compensation	The relative positions of the centers of the six cameras' CMOS and the phase center of the onboard D-RTK antenna have been calibrated and are recorded in the EXIF data of each image.



• GNSS

Single-Frequency High-Sensitivity GNSS GPS + BeiDou + Galileo²⁾ (Asia); GPS + GLONASS + Galileo²⁾ (other regions)

Multi-Frequency Multi-System High-Precision RTK GNSS

Frequency Used

GPS: L1/L2; GLONASS: L1/L2; BeiDou: B1/B2; Galileo^[2]: E1/E5

First-Fixed Time: < 50 s

Positioning Accuracy: Vertical 1.5 cm + 1 ppm (RMS); Horizontal 1 cm + 1 ppm (RMS). 1 ppm indicates error with a 1 mm increase over 1 km of movement.

Velocity Accuracy: 0.03 m/s

• Mapping Functions

Ground Sample Distance (GSD)

(H/18.9) cm/pixel, H indicates the aircraft altitude relative to the area mapped (unit: m)

Rate of Data Collection

Max operating area of approx. 0.63 km² for a single flight at an altitude of 180 m, i.e., GSD is approx. 9.52 cm/pixel, with a forward overlap rate of 80% and a side overlap ratio of 60%, during a flight that drains the battery from 100% to 30%.

• Gimbal

Controllable Range

Pitch: -90° to +30°

• Vision System

Velocity Range

≤ 31 mph (50 kph) at 6.6 ft (2 m) above ground with adequate lighting

Altitude Range

0 - 33 ft (0 - 10 m)

Operating Range

0 - 33 ft (0 - 10 m)

Obstacle Sensory Range

2 - 98 ft (0.7 - 30 m)

Operating Environment

Surfaces with clear patterns and adequate lighting (> 15 lux)

• Camera

Sensors

Six 1/2.9" CMOS, including one RGB sensor for visible light imaging and five monochrome sensors for multispectral imaging.

Each Sensor: Effective pixels 2.08 MP (2.12 MP in total)

Filters

Blue (B): 450 nm ± 16 nm; Green (G): 560 nm ± 16 nm; Red (R): 650 nm ± 16 nm; Red edge (RE): 730 nm ± 16 nm; Near-infrared (NIR): 840 nm ± 26 nm

Lenses

FOV (Field of View): 62.7°

Focal Length: 5.74 mm (35 mm format equivalent: 40 mm), autofocus set at ∞

Aperture: f/2.2

RGB Sensor ISO Range

200 - 800

Monochrome Sensor Gain

1 - 8x

Electronic Global Shutter

1/100 - 1/20000 s (visible light imaging); 1/100 - 1/10000 s (multispectral imaging)

Max Image Size

1600×1300 (4:3.25)

Photo Format

JPEG (visible light imaging) + TIFF (multispectral imaging)

Supported File Systems

FAT32 (≤ 32 GB); exFAT (> 32 GB)

Supported SD Cards

microSD with a minimum write speed of 15 MB/s. Max Capacity: 128 GB. Class 10 or UHS-I rating required

Operating Temperature

0° to 40° C (32° to 104° F)

• Remote Controller

Operating Frequency

2.4000 GHz to 2.4835 GHz (Europe, Japan, Korea)

5.725 GHz to 5.850 GHz (Other countries/regions)^[1]

Transmission Power (EIRP)

2.4 GHz: < 20 dBm (CE / MIC / KCC)

5.8 GHz: < 26 dBm (FCC / SRRC / NCC)

Max Transmission Distance

FCC / NCC: 4.3 mi (7 km); CE / MIC / KCC / SRRC: 3.1 mi (5 km)

(Unobstructed, free of interference)

Built-in Battery

6000 mAh LiPo 2S

Operating Current / Voltage

1.2 A @ 7.4 V

Mobile Device Holder

Tablets and smartphones

Operating Temperature

0° to 40° C (32° to 104° F)

• Intelligent Flight Battery (PH4-5870mAh-15.2V)

Capacity

5870 mAh

Voltage

15.2 V

Battery Type

LiPo 4S

Energy

89.2 Wh

Net Weight

468 g

Operating Temperature

-10° to 40° C (14° to 104° F)

Charging Temperature

5° to 40° C (41° to 104° F)

Max Charging Power

160 W

• Intelligent Flight Battery Charging Hub (P4CH)

Voltage

17.5 V

Operating Temperature

5° to 40° C (41° to 104° F)

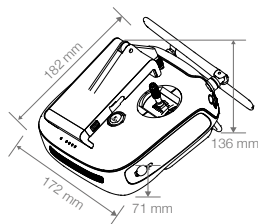
• AC Power Adapter (PH4C160)

Voltage

17.4 V

Rated Power

160 W



※ This Quick Start Guide is subject to change without prior notice.

Download the latest version from

<https://www.dji.com/p4-multispectral>

[1] To comply with local laws and regulations, this frequency is not available in some countries or regions.

[2] Support for Galileo is coming soon.